

DTU



Model-based Machine Learning

Mixture models

Mixture models

Road to MBML: where are we?

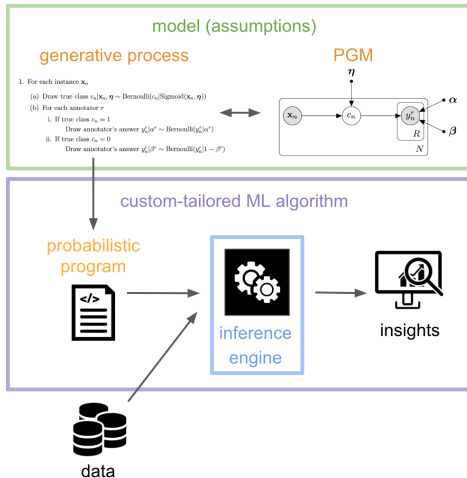
Foundations of PGMs and probabilistic programming

Probabilistic modelling in various contexts - your modelling toolbox

Bayesian inference (exact and approximate)

Week	Topic
1	Intro to the course + Prob. review
2	PGM foundations
3	PGM foundations II
4	Freq. vs Bayesian + Prob. Prog. + Mixture models
5	Regression models
6	Classification and Hierarchical models
7	Temporal models
8	Topic Models
9	Markov-chain Monte Carlo (MCMC)
10	Variational inference
11	Generative models
12	Gaussian processes
13	Project support

Model-based Machine Learning



At the end of this lecture, you should be able to:

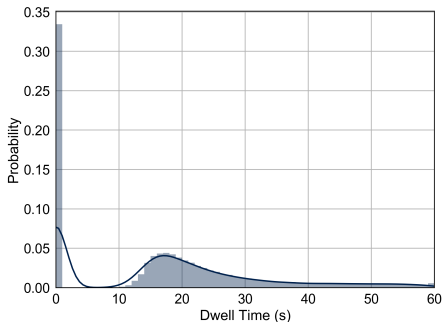
- Understand the concept mixture model, its representation, and inference challenges
- Implement mixture models in Pyro and apply them to model data

Mixture models

- A PGM is composed of observed and latent variables, parameters, constants.
- In this course, we'll approach some examples from this very large family
- Mixture models are pervasive in data modelling in general

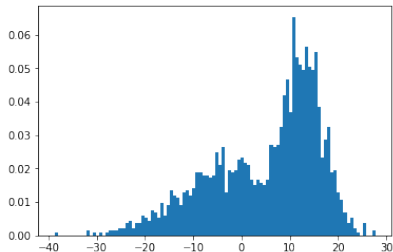
Mixture models

- A PGM is composed of observed and latent variables, parameters, constants.
- In this course, we'll approach some examples from this very large family
- Mixture models are pervasive in data modelling in general
- Problem:
 - Sub-populations of data
 - Data generated from combination/competition of multiple sources
 - Number of sources usually discrete and finite



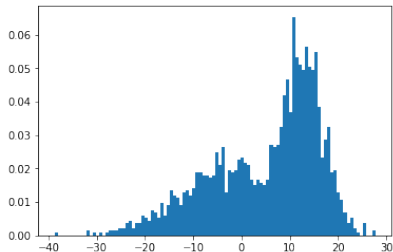
The canonical example: Gaussian Mixture

- What we observe

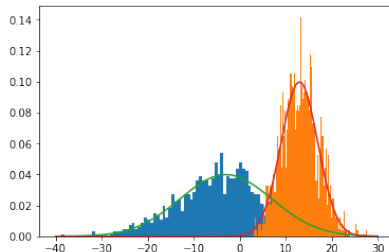


The canonical example: Gaussian Mixture

- What we observe



- What really happens

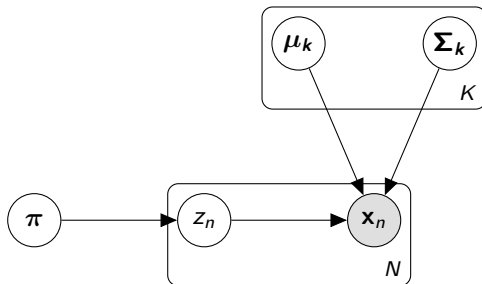


Generative story

- ① Draw π , and (μ_k, Σ_k) for all K gaussians
- ② For $n = 1, 2, \dots, N$
 - ① Draw $z_n \sim \text{Categorical}(\pi)$
where π is a vector $(1 \times K)$ with the probabilities of each class
 - ② Generate $\mathbf{x}_n$, from the k -th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\mu_{z_n}, \Sigma_{z_n})$

Generative story

- 1 Draw π , and (μ_k, Σ_k) for all K gaussians
- 2 For $n = 1, 2, \dots, N$
 - 1 Draw $z_n \sim \text{Categorical}(\pi)$
where π is a vector $(1 \times K)$ with the probabilities of each class
 - 2 Generate $\mathbf{x}_n$, from the k -th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\mu_{z_n}, \Sigma_{z_n})$



- ① Draw π
- ② Draw (μ_k, Σ_k) for all K Gaussians
- ③ For $n = 1, 2, \dots, N$
 - ① Draw $z_n \sim \text{Categorical}(\pi)$
where π is a vector $(1 \times K)$ with the probabilities of each class
 - ② Generate $\mathbf{x}_n$, from the k -th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\mu_{z_n}, \Sigma_{z_n})$

Factorization:

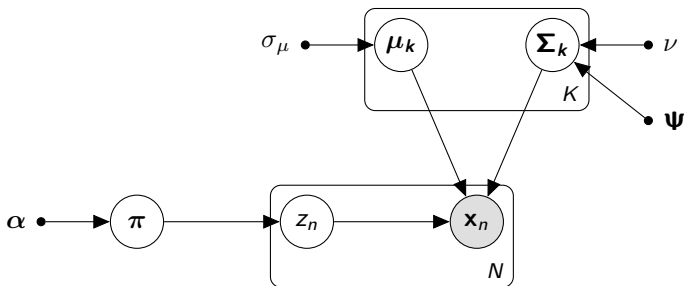
$$p(\pi) \left(\prod_{k=1}^K p(\mu_k) p(\Sigma_k) \right) \prod_{n=1}^N p(z_n | \pi) p(\mathbf{x}_n | \mu_{z_n}, \Sigma_{z_n})$$

Note: in practice we need to be exhaustive

...particularly in probabilistic programming (e.g. Pyro)

- $\pi \sim \text{Dirichlet}(\alpha)$
- $\mu_k \sim \mathcal{N}(\mathbf{0}, \sigma_\mu \mathbf{I})$
- $\Sigma_k \sim \mathcal{W}^{-1}(\Psi, \nu)$

$\mathcal{W}^{-1}$ denotes the Inverse Wishart distribution; typically ν = degrees of freedom (typically number of dimensions of $\mathbf{x}$), and $\Psi = \mathbf{I}$



Note: in practice we need to be exhaustive

...particularly in probabilistic programming (e.g. Pyro)

- $\pi \sim \text{Dirichlet}(\alpha)$
- $\mu_k \sim \mathcal{N}(\mathbf{0}, \sigma_\mu \mathbf{I})$
- $\Sigma_k \sim \mathcal{W}^{-1}(\Psi, \nu)$
 $\mathcal{W}^{-1}$ denotes the Inverse Wishart distribution; typically $\nu = \text{degrees of freedom}$ (typically number of dimensions of $\mathbf{x}$), and $\Psi = \mathbf{I}$

The factorization becomes:

$$\text{Dirichlet}(\pi|\alpha) \left(\prod_{k=1}^K \mathcal{N}(\mu_k|\mathbf{0}, \sigma_\mu \mathbf{I}) \mathcal{W}^{-1}(\Sigma_k|\Psi, \nu) \right) \prod_{n=1}^N \pi_{z_n} \mathcal{N}(\mathbf{x}|\mu_{z_n}, \Sigma_{z_n})$$

Full generative process

- ① Draw mixture components $\pi \sim \text{Dirichlet}(\alpha)$
- ② For each mixture component (Gaussian) $k \in \{1, \dots, K\}$
 - ① Draw mean vector $\mu_k \sim \mathcal{N}(\mathbf{0}, \sigma_\mu \mathbf{I})$
 - ② Draw covariance matrix $\Sigma_k \sim \mathcal{W}^{-1}(\Psi, \nu)$
- ③ For $n = 1, 2, \dots, N$
 - ① Draw $z_n \sim \text{Categorical}(\pi)$
where π is a vector ($1 \times K$) with the probabilities of each class
 - ② Generate $\mathbf{x}_n$, from the k -th Gaussian, $\mathbf{x}_n \sim \mathcal{N}(\mu_{z_n}, \Sigma_{z_n})$

Note that the concept of mixture models **can be extended to other distributions beyond Gaussians!**

- Gaussian mixture models - “K-means clustering” (Part 3 of the notebook)
 - See “03 - Probabilistic Programming with Pyro.ipynb” notebook
 - Expected duration: 30 minutes

- **Main reading:** Chapter 9.2: “Mixture Models and EM”, pages 430-435 of Chris Bishop’s book, “Pattern Recognition and Machine Learning” (PRML)
URL: <https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf>)
- More on Mixture Models: Chapter 11: “Mixture models and the EM algorithm”, pages 337-345 of Kevin Murphy’s book “Machine Learning: A Probabilistic Perspective”